

Dealer Gamma Exposure (GEX) & Gamma-Flip Engine

Complete technical reference: architecture, quantitative methodology, every module and function, production-hardening history, and interview preparation.

Component	File	LOC	Responsibility
Quant + CLI core	gex.py	2,179	BSM gamma, dual profiles, GEX aggregation, flip search + time decay, walls, Schwab data layer, token persistence, rendering, charting, CLI
Test suite	test_gex.py	1,027	53 pytest cases: closed-form math validation, parser/filter behaviour, profile weighting, token plumbing
HTTP API	server.py	480	FastAPI JSON endpoints, caching, client lifecycle, phone re-auth
OI archive analysis	scripts/oi_history.py	371	dOI history, aggressor lean, empirical dealer sign from CBOE open-close
Mobile app	static/index.html	308	Installable iOS PWA: SPY/QQQ, expiry picker, live levels, canvas chart
Token forensics	scripts/token_audit.py	231	Diagnoses early token expiry from the token journal
OAuth setup	scripts/schwab_setup.py	165	Browser login, token validation, live verification pull
Phone re-auth UI	static/setup.html	155	Weekly Schwab re-login without a terminal
Push notifier	scripts/daily_gex.sh	110	Multi-cadence runner, message assembly, ntfy/Pushover/Telegram
Schedulers	scripts/*.plist	4 files	launchd agents: daily/weekly/monthly pushes + always-on API server

Stack: Python 3.9+ core (3.10+ for live data) · NumPy · SciPy · Matplotlib · FastAPI · Uvicorn · schwab-py (OAuth 2.0) · pytest · launchd · HTML/CSS/JS PWA

Data source: Charles Schwab Trader API (Market Data). **Instruments:** SPY, QQQ, and any optionable underlying with listed open interest. **Total:** ~5,000 lines.

1. Executive Summary

This project estimates **dealer gamma exposure** (GEX) from live options-chain data and derives the price levels where market-maker hedging flow changes character. Those levels form a daily trading bias: where volatility is likely to be suppressed, where it is likely to be amplified, and which strikes act as magnets or accelerants.

The 30-second pitch

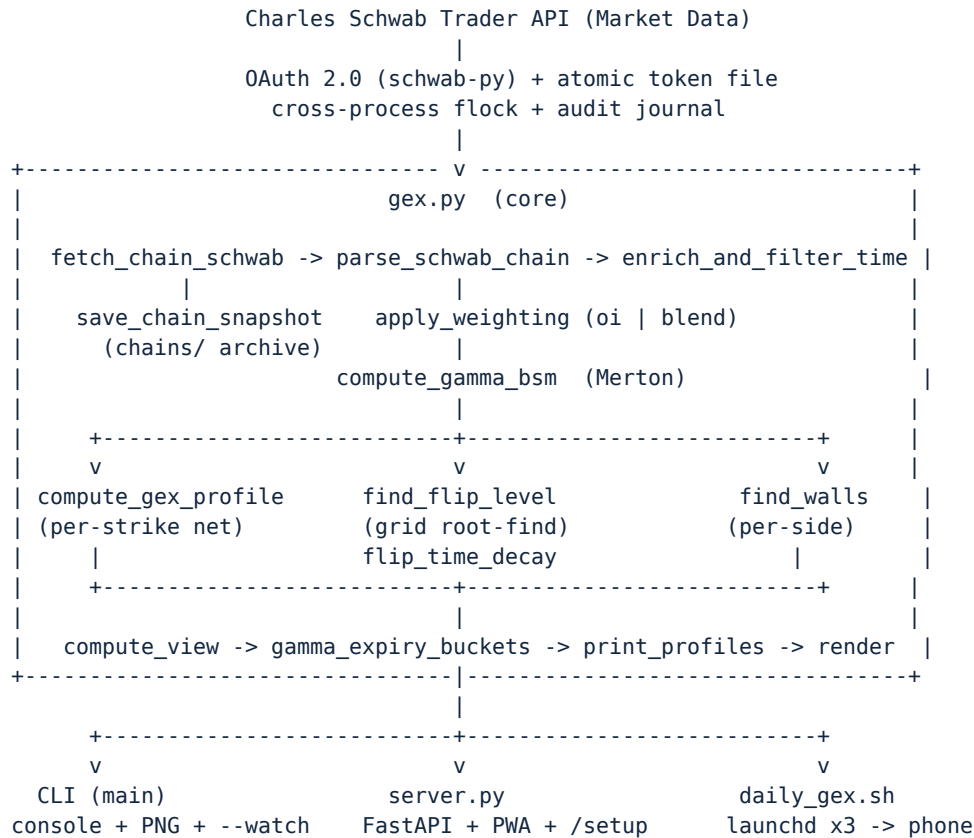
Options market makers hold inventory whose delta changes as spot moves. To stay hedged they must trade the underlying continuously, and the *direction* of that forced trading depends on the sign of their aggregate gamma. When dealers are **long gamma** they sell rallies and buy dips, damping volatility. When dealers are **short gamma** they buy rallies and sell dips, amplifying it. The system computes where that sign flips and turns it into an actionable pre-market briefing delivered to a phone.

What separates this from a retail GEX chart

- **Gamma is recomputed from scratch** with Black-Scholes-Merton on **per-strike** implied volatility. Vendor greeks are never trusted and the smile is never flattened to one ATM vol.
- **Two profiles, not one.** An intraday (trading) layer blends today's volume into stale open interest for near-dated contracts; a structural (overview) layer uses durable OI only. Conflating them is a modeling error.
- **The flip is found by root-finding** across a grid of hypothetical spot prices, then **projected forward in time** to where it migrates by the 16:00 close.
- **The core assumption is stress-tested every run**, and can now be **empirically tested** against CBOE open-close data rather than merely assumed.
- **It runs unattended in production**, hardened against real failures observed live: OAuth token revocation, trickling network responses, vendor data gaps, macOS permission walls.

2. Architecture and Data Flow

A single computational core is consumed by three independent delivery paths. No math is duplicated: the API, the scheduled notifier, and the CLI all call the same audited functions.



Design principle: one source of truth

Every delivery path is a thin adapter. `server.py` imports `gex` and calls the same functions the CLI does; the notifier shells out to the CLI. A change to the gamma formula propagates everywhere at once, and the test suite covers all three paths by covering the core.

3. The Quantitative Core

3.1 Gamma per contract (Merton closed form)

```

gamma = exp(-q*T) * phi(d1) / (S * sigma * sqrt(T))
d1    = [ ln(S/K) + (r - q + sigma^2 / 2) * T ] / (sigma * sqrt(T))
phi(x) = exp(-x^2 / 2) / sqrt(2*pi)
  
```

Symbol	Meaning	Source in this system
S	Spot price	underlyingPrice from the chain snapshot (live)
K	Strike	Per contract
T	Time to expiry (years)	Calendar ACT/365 to 16:00 ET, floored at 300 s
σ (sigma)	Implied volatility	Per-strike vendor IV, percent → decimal

Symbol	Meaning	Source in this system
r	Risk-free rate	--rate, default 0.0469 , always printed
q	Dividend yield	Per-ticker map (SPY 1.2%, QQQ 0.6%), overridable

The $\exp(-q \cdot T)$ factor. The dividend-adjusted (Merton) form carries an explicit discount factor that plain Black-Scholes gamma omits. This was a genuine bug found in review: q appeared inside d_1 but not as the leading factor, overstating gamma by roughly $q \cdot T$ whenever a dividend yield was supplied. Silent at $q=0$, wrong the moment a real yield was passed.

Implementation notes

- **Fully vectorized.** S, K, T and σ broadcast as NumPy arrays, pricing the whole chain in one operation rather than a Python loop over thousands of contracts.
- **Degenerate inputs return 0, never NaN.** A validity mask ($T > 0$, $\sigma > 0$, $S > 0$, $K > 0$) means a sparse or malformed chain cannot poison aggregate levels with NaN or infinity.
- **Custom normal density.** `scipy.stats.norm.pdf` carries heavy per-call overhead; the flip search evaluates ϕ tens of millions of times per run. Replacing it with a direct NumPy expression cut runtime $\sim 15\times$ (60 s $\rightarrow$ 4 s per ticker).

3.2 Exposure weighting: the two profiles

The same contract set answers two different questions, and conflating them is a modeling error.

	INTRADAY (trading layer)	STRUCTURAL (overview layer)
Scope	0DTE $\rightarrow$ front week + next monthly OpEx	Every expiry in the fetched window
Weighting	OI blended with today's volume	Open interest only
Rationale	That is where essentially all real-time hedging pressure lives; prior-close OI is stale by mid-morning on 0DTE	Durable positioning, not today's churn
Behaviour	Moves continuously — recompute on a fast cadence (--watch)	Barely moves day to day; that is the point
Answers	Today's tradeable flip and nearest walls	Multi-day walls and the regime flip

```
weight(contract) =
    oi                                # structural
    oi + volume * w(DTE)             # intraday
    w = 1.0    for 0DTE
    w = 0.5    through the front week (<= 5 days)
    w = 0.0    beyond
```

Stated assumption. Traded volume includes *closing* as well as opening trades, so the blend is an **upper bound** on fresh positioning. It deliberately biases toward over-counting near-dated activity, because understating live 0DTE gamma is the larger error. Every run prints the **volume uplift** versus OI-only (a recent live QQQ run: **+159.5%**), so the size of the correction is always visible rather than hidden.

3.3 Dollar gamma

dollar GEX per contract = $\text{gamma} * \text{weight} * \text{multiplier} * S^2 * 0.01$

Unit convention (a real category error). This is the **per-1%** convention. Per-**point** GEX is $\text{gamma} \times \text{weight} \times \text{multiplier} \times S$, equal to the per-1% figure divided by $(0.01 \times S)$. SqueezeMetrics publishes per-point; most retail charts publish per-1%. Comparing them raw is meaningless. The tool prints which convention it uses on every run.

3.4 Dealer sign convention — the biggest assumption

net GEX = SUM(call GEX) - SUM(put GEX) # dealers long calls, short puts

Exposed as configuration and flippable from the CLI. Historically this could only be assumed; it can now be **empirically tested** — see §3.8.

3.5 The gamma flip: root-finding, then time decay

```
for each hypothetical spot S' in linspace(spot*0.90, spot*1.10, 1000):
    recompute gamma(S') for EVERY contract      # K, T, sigma, weight fixed
    curve[i] = sum( sign * gamma(S') * weight * mult * S'^2 * 0.01 )
```

interpolate each sign change: $x = x_0 - y_0 * (x_1 - x_0) / (y_1 - y_0)$

- **Both S-dependent terms are recomputed** — gamma itself and the S^2 dollar factor.
- **Multiple crossings are detected, not assumed away.** All are collected; the nearest to spot is reported and the rest listed.
- **Linear interpolation, not nearest-node**, so precision does not depend on grid resolution alone.
- **Time decay projection.** `flip_time_decay()` re-runs the search with every contract's T advanced to the 16:00 ET close, dropping contracts that expire inside the window. The flip is not static: it migrates through the session as gamma decays, and the projection shows where it is heading.

Sticky-strike caveat. Holding each contract's IV fixed while spot moves is the sticky-strike assumption. Reality sits between sticky-strike and sticky-delta. The flip is an estimate under a stated volatility-dynamics assumption, not a model-free level.

3.6 Walls — computed per side, not from the net

A subtle correction over the naive definition. Net GEX at a strike is $(\text{call_gex} - \text{put_gex})$, so a strike carrying **both** heavy call and heavy put gamma nets to ≈ 0 and would be invisible to a net-based search — even though it is a major hedging battleground. Therefore:

Output	Definition	Trading meaning
Call wall	Strike of MAX call_gex (per-side)	Resistance / pin
Put wall	Strike of MIN put_gex (most negative, per-side)	Support that becomes a downside accelerant once breached
Regime	spot > flip → LONG gamma; spot < flip → SHORT	Long: vol-damping, mean-reverting. Short: vol-amplifying, trend-prone.

3.7 Uncertainty quantification

Probe	Method	Interpretation
Literal sign flip	Recompute the flip with put_sign inverted	Usually makes <i>all</i> gamma positive so the flip vanishes — a flip exists <i>only</i> under the short-put assumption
Graded band	Recompute at $\pm 50\%$ of the short-put magnitude	An actual distance: how far the level moves under plausible mis-specification
Layer agreement	Intraday flip vs structural flip	Divergence $> 1\%$ of spot means near-dated flow is pulling the flip off the standing structure

A **LOW CONFIDENCE** warning prints when the flip moves more than **1% of spot** across the band, and surfaces as a badge in the mobile app. A live example: SPY flip 758.35 with a band of 741.03–768.26 (2.25% of spot) — but spot sat *outside* the entire band, so the **regime call was robust even though the level was not**. That distinction is what matters for trading.

3.8 Empirical dealer sign (CBOE open-close)

The model's central assumption can now be tested rather than asserted. `scripts/oi_history.py --open-close` ingests a CBOE Open-Close CSV (customer opening buy/sell per strike) and computes the sign implied by actual customer flow:

```
net_customer_opening = cust_open_buy - cust_open_sell
```

```
net_customer_put BUYING > 0 -> dealers SHORT puts (put_sign -1: assumption HOLDS)
net_customer_put SELLING > 0 -> dealers LONG puts (put_sign +1: assumption FAILS)
```

Customers and dealers sit on opposite sides of opening customer flow, so the inversion is direct. The report states plainly whether the data supports or contradicts the modelled convention.

3.9 Hedging-urgency decomposition

The governing discipline: **match the expiry set to the holding period**. Every run prints a first-match bucket table so the headline number is never mistaken for uniformly actionable gamma:

Bucket	Why it matters
0DTE (expires today)	Enormous intraday gamma that evaporates at 16:00 ET ; also where prior-close OI is least trustworthy
≤ 1 week	Fast money, genuinely reheded on an intraday horizon
$\leq$ monthly OpEx (3rd Friday)	Holds the bulk of index OI. Charm and vanna dominate into roll-off (not modelled)
Beyond OpEx	Slow money — will not be rebalanced today; inflates headline numbers without driving today's flow

The intraday profile additionally prints a **0DTE decay cliff** warning: what share of that profile expires at the close and how long remains.

4. Module Reference: gex.py

2,179 lines in labelled sections. Private helpers are prefixed with an underscore.

4.1 Data model

Object	Purpose
Contract	strike, expiry, type, oi, iv, T, volume (today's, live), and size — the exposure weight actually used, stamped by <code>apply_weighting()</code>
DealerConvention	call_sign, put_sign, label. Makes the central assumption a first-class swappable object rather than a magic number
Config	ticker, multiplier, rate, div_yield (+ provenance), price_range, steps, both conventions. Threaded everywhere so nothing reads globals

4.2 Profiles, pricing and aggregation

Function	Contract
contract_weight / apply_weighting	Per-contract exposure weight: OI, or OI blended with today's volume by DTE tier
select_profile_contracts	Filters to a profile's expiry scope and stamps its weighting. Returns copies , so the two profiles never share mutated state
intraday_horizon	max(front week, next monthly OpEx)
zero_dte_cliff	Share of a profile expiring at today's close, plus time remaining
_norm_pdf	Vectorized normal density; ~15× faster than scipy in the hot path
compute_gamma_bsm	Merton gamma, broadcastable, 0 for degenerate inputs. The most tested function in the codebase
_to_arrays / _signed_dollar_gex	Packs contracts into parallel arrays (using size, not raw oi) and applies the dealer sign
compute_gex_profile	Scatter-add aggregation (np.unique + searchsorted + np.add.at) onto the sorted strike axis; keeps call, put and net vectors
gross_dollar_gamma	Sign-agnostic Σ dollar gamma for concentration and bucket shares

4.3 Flip, decay and walls

Function	Contract
find_flip_level	Grid root-finder. Returns nearest crossing, <i>all</i> crossings, the (grid, curve) arrays, and total GEX at spot
_decayed_contracts / flip_time_decay	Projects the flip to the 16:00 ET close by advancing every T and dropping contracts that expire inside the window
_flip_with_put_sign	Helper for the graded sensitivity band

Function	Contract
find_walls	Per-side extremes (max call_gex / min put_gex), not net — catches strikes with heavy gamma on both sides
compute_view	Orchestrator: profile + walls + flip + flipped flip + sensitivity band + gross gamma
print_profiles	Renders both layers, the volume uplift, the decay cliff, and the layer-agreement verdict

4.4 Time handling

Function	Contract
_et_tz / now_et	America/New_York via zoneinfo, so DST transitions are handled correctly
seconds_to_expiry	Seconds to 16:00 ET on the expiration date (PM settlement at the cash close)
enrich_and_filter_time	Fills T, drops settled contracts, floors T at 300 s; returns both counts
third_friday / next_monthly_opex	Monthly OpEx calendar arithmetic including year rollover
explain_empty_view	Explains WHY a view is empty — 'today's ODTE settled at 16:00 ET (7h 20m ago) — expired, not missing' beats a bare '(empty)'

Why the T floor exists. Gamma carries $S \cdot \sigma \cdot \sqrt{T}$ in the denominator, so as $T \rightarrow 0$ at-the-money gamma diverges. Without a floor a ODTE chain minutes before the close produces meaningless astronomical levels. Clamped contracts are counted and warned about.

4.5 Data layer, token safety and archive

Function	Contract
get_schwab_client	Builds a schwab-py client via client_from_access_functions with OUR read/write funcs. Lazy import (schwab-py needs 3.10+). Stamps token path, mtime and credentials onto the client
_token_reader / _token_writer	Journals every touch; writes ATOMICALLY (temp file + fsync + os.replace) so a crash or overlap can never leave a truncated token
token_audit / _token_fingerprint	Forensic journal with PID, command and a sha256 prefix of the refresh token — rotation is visible, the secret never is
token_lock	Cross-process flock on a sidecar file, re-entrant within a process; degrades to unlocked on timeout (a race beats an outage)
reload_client_if_stale	Rebuilds from disk if the token changed. Must run INSIDE the lock — checking earlier is a check-then-act race
fetch_chain_schwab	One un-paginated chain request; retries transient faults, raises immediately on auth/4xx
parse_schwab_chain	Walks the nested expiry→strike→contract maps, applies every data-quality filter, captures volume

Function	Contract
save_chain_snapshot	Archives the RAW payload (pre-filter) to chains/<TICKER>/<date>.csv.gz — open interest is never backfillable
fetch_spx_spy_ratio	Live SPX/SPY ratio computable from either leg

Data-quality filters

Filter	Rule	Rationale
Zero open interest	Drop $OI \leq 0$	GEX is OI-weighted
Missing / sentinel IV	Drop non-finite or ≤ 0 (catches -999.0)	Garbage IV in, garbage gamma out
Crossed quotes	Drop bid > ask	Stale or locked market
Deep-ITM bad quotes	Drop if >5% ITM and (no bid or rel. spread >25%)	Deep-ITM IV comes from a sliver of extrinsic value and is noise-dominated
OTM wings	Never quote-filtered	Zero-bid wings still carry real tail gamma; filtering them would destroy wing gamma

5. Delivery Layer

5.1 CLI

Flag	Effect
--ticker / --expiry / --all-days	Underlying; 0dte all YYYY-MM-DD; expiration window (default 45)
--profiles	Show the INTRADAY and STRUCTURAL layers side by side
--watch N	Re-fetch every N seconds, printing spot/flip/walls and deltas per tick
--rate / --div-yield	Override r and q; both echoed with provenance
--convention / --call-sign / --put-sign	Flip or hand-set the dealer sign assumption
--price-range / --steps	Flip-search window ($\pm 10\%$) and grid resolution (1000)
--levels-only	Compact output for notifications
--no-save-chain / --chain-dir	Archive controls (archiving is ON by default)
--demo	Deterministic synthetic chain; no credentials, no network

5.2 server.py — FastAPI JSON API

Endpoint	Returns
GET /api/health	Token age, days remaining, and the last real auth outcome — file age alone lies when a token family has been revoked
GET /api/expirations	Listed expirations with already-settled dates excluded, so an empty-chain selection is impossible
GET /api/gex	Spot, per-view levels (flip, crossings, walls, regime, interpretation, low-confidence), per-strike profile, expiry buckets, drop counts, warnings. fresh=1 bypasses the 60 s cache
GET /api/auth/start	Schwab authorize URL for phone-based re-login
POST /api/auth/complete	Exchanges the pasted redirect URL, writes the token under the lock, validates a refresh token is present

5.3 Mobile app and notifications

- **PWA** (static/index.html): SPY/QQQ toggle, live expiration picker, a single **GET FRESH LEVELS** button that bypasses the cache, regime banner, level cards, hand-drawn canvas chart with DPR scaling, expiry-bucket table. Installs to the home screen; no build step, no external dependencies.
- **Phone re-auth** (static/setup.html): the weekly Schwab login without a terminal. The header token indicator turns amber inside the last 1.5 days and links here.

- **Notifications:** three launchd cadences — daily (Mon-Fri 07:45 CT, 45-day window), weekly (Monday, 90 days), monthly (1st, 150 days) — for SPY and QQQ, via ntfy / Pushover / Telegram. Failure is itself a notification: PARTIAL, RUN FAILED, and SCHWAB TOKEN EXPIRED carry the fix.
- **Chain archive:** every live fetch writes chains/<TICKER>/<date>.csv.gz (~95 KB/ticker/day, ~49 MB/year for two). Analysed by scripts/oi_history.py for dOI, aggressor lean and empirical dealer sign.

6. Assumptions, Limitations and Deliberate Deviations

6.1 Standing assumptions

Assumption	Status and impact
Dealer sign convention	The biggest weakness. The flip's very existence depends on it. Mitigated by the sensitivity band, and now testable against CBOE open-close data
Open interest is prior-session	The freshest OI available is yesterday's close. Mitigated intraday by blending today's volume in the intraday profile
Volume blend upper bound	Volume includes closing trades, so the blend over-counts fresh positioning — a deliberate bias, since understating live ODTE gamma is worse
Sticky-strike repricing	Per-strike IV held fixed across the flip grid; reality is between sticky-strike and sticky-delta
European exercise	BSM gamma for American ETF options. The early-exercise premium concentrates in deep-ITM (low-gamma) strikes
Charm and vanna ignored	Second-order flows dominate into monthly OpEx roll-off and are explicitly out of scope

6.2 Deliberate deviations from the textbook pipeline

Calendar time, not trading time

A common prescription measures ODTE T in trading minutes (390×252). This system uses calendar ACT/365 because gamma depends on $\sigma^2 \cdot T$ and the input IVs are **vendor quotes annualized on a calendar clock**. Pairing a calendar-annualized sigma with a trading-time T breaks that pairing and inflates ODTE gamma severalfold. A trading-time clock is correct only when IV is re-derived under the same clock. Internal consistency beats superficial sophistication.

Single underlying, not the full index complex

A complete S&P dealer book would aggregate SPX + SPXW + XSP + SPY + ES normalized to common notional. Impossible here: Schwab returns **zero open interest for cash-index options** and carries no futures options. The tool measures one listed underlying as a correlated proxy, states the limit explicitly, and **refuses to sum raw GEX across underlyings** — which without notional normalization would be meaningless.

6.3 Hard data-source constraints, discovered empirically

Constraint	Evidence and consequence
No index open interest	Verified live: every \$SPX strike returns OI = 0 while SPY returns full OI (max ~37k). Index GEX is impossible on Schwab. Default ticker changed to SPY

Constraint	Evidence and consequence
Chain size limits	A full-year SPX chain returns HTTP 502; default window reduced to 45 days with a 502-specific hint
7-day refresh token	Schwab rejects refresh tokens older than seven days with no programmatic renewal. Surfaced proactively from day 5.5
No dealer-direction feed	Schwab carries no open-close volume, so the sign convention needs an external CSV to be validated

7. Testing Strategy

53 pytest cases, no network access required, full suite under 5 seconds. The philosophy: **financial code must be validated against independently derived truth**, not against its own output.

7.1 Mathematical validation

Test	Reference value and derivation
ATM gamma, clean case	$S=K=100, T=1, \sigma=0.2, r=q=0$ gives $d_1=0.1$ analytically; expected 0.0198476275, asserted to 1e-9
Hull textbook example	$S=49, K=50, r=0.05, \sigma=0.20, T=20/52 \rightarrow$ gamma ≈ 0.066 as published in Hull. An external reference, not self-generated
Dividend discount	Constructed so $d_1 = 0$ exactly ($r=0, q=0.02, \sigma^2/2$ cancels), isolating $\exp(-q \cdot T)$; asserted to 1e-12
Degenerate inputs	$T \leq 0, \sigma \leq 0, S \leq 0$ must return exactly 0, never NaN or infinity

7.2 The flip test: a hand-derived analytical crossing

The most valuable test in the suite. For one call at K_c and one put at K_p with equal weight and identical σ and T , the shared $1/(S \cdot \sigma \cdot \sqrt{T})$ and S^2 factors cancel at the zero, so the crossing occurs where $\phi(d_1^c) = \phi(d_1^p)$, i.e. $d_1^c = -d_1^p$:

$$S^* = \sqrt{(K_c \cdot K_p)} \cdot \exp(-c \cdot T), \quad c = r - q + \sigma^2 / 2$$

$$K_c = 90, K_p = 110, \sigma = 0.2, T = 1, r = q = 0 \rightarrow c = 0.02$$

$$S^* = \sqrt{9900} \cdot \exp(-0.02) = 97.5288$$

The grid root-finder must land on 97.5288 within 0.05. This validates the interpolation, the sign-change detection and the nearest-crossing selection against mathematics rather than a golden output file. Companion tests assert no crossing exists when all gamma is positive, and that the flip is invariant to multiplier and weight scaling.

7.3 Behavioural and infrastructure tests

Area	What is asserted
Profile weighting	DTE-tiered volume blend (1.0 / 0.5 / 0.0); structural ignores volume; profiles return copies so they cannot share mutated state
Per-side walls	A strike with huge call <i>and</i> put gamma (net ≈ 0) is still found as both walls — the regression the net-based definition missed
Flip time decay	Returns current and decayed flip, drops contracts expiring inside the window, floors tiny T consistently
0DTE decay cliff	Share and time-remaining computed; returns None when no 0DTE is present
Chain parsing / filters	Hand-built payload in Schwab's nested shape; crossed and deep-ITM quotes dropped, deep-OTM zero-bid wing preserved
Empirical dealer sign	Open-close CSV parsing, the customer $\Rightarrow$ dealer inversion, and the supported/fails verdict in both directions

Area	What is asserted
Empty-view messaging	'Expired' vs 'missing' distinguished by clock and weekday, with injected times so it never depends on when tests run
Token persistence	Writes are atomic (a failed write leaves the previous token intact); the journal shows rotation via fingerprints; raw tokens never appear in the log
Locking	A separate process is genuinely blocked while the lock is held; nested acquisition does not deadlock; timeout degrades to unlocked
Retry classification	Transient timeouts retry then succeed; OAuth and 4xx raise immediately without retrying

8. Production Hardening: Bugs Found and Fixed

Every item below was discovered by running the system against live markets. Each has a symptom, a root cause and a verified fix — the strongest interview material in the project.

8.1 Quantitative correctness

Defect	Root cause	Fix
Gamma overstated when a dividend yield was supplied	The Merton $\exp(-q \cdot T)$ leading factor was missing; q appeared only inside d_1	Added the factor; regression test with $d_1=0$ isolates it exactly
Walls missed contested strikes	Walls were taken from NET GEX, so a strike with heavy call and put gamma netted to ≈ 0 and vanished	Per-side extremes: <code>max call_gex / min put_gex</code>
SPX-equivalent off by ~ 25 points	Cross-quote used a hardcoded 10.0; the live ratio was 10.034	Live ratio fetched from either leg, source label printed
Flip reported as 'not found'	$\pm 5\%$ search window too narrow in deeply short-gamma sessions	Widened to $\pm 10\%$
Intraday gamma understated	Prior-close OI is stale by mid-morning on ODTE	Intraday profile blends today's volume by DTE tier; uplift printed

8.2 Performance

Symptom: a two-ticker scheduled run took roughly two minutes. **Diagnosis:** the flip search evaluates the normal PDF across 1,000 grid nodes $\times$ thousands of contracts — tens of millions of calls per run — and `scipy.stats.norm.pdf` carries substantial per-call overhead. **Fix:** a direct NumPy density expression. **Result:** $60\text{ s} \rightarrow 4\text{ s}$ per ticker ($\sim 15\times$), whole pipeline 8 s, identical numerical output verified by the existing tests.

8.3 Reliability and distributed systems

Defect	Root cause	Fix
A single request hung for 868 seconds	httpx read-timeouts fire only on socket silence; a response trickling bytes slowly never triggers one	Hard wall-clock deadline via ThreadPoolExecutor, independent of socket activity
Scheduled runs died on transient hiccups	Any exception aborted the ticker	Retry with linear backoff for transient faults; immediate raise for auth and 4xx
Scheduled jobs silently never executed	macOS TCC: launchd's bash had no Full Disk Access to the project directory	Diagnosed via launchd stderr; documented the grant and verified by kickstart
A login 'succeeded' but wrote an unusable token	A bad authorization code yields a degenerate short-lived token with no refresh token	Post-login validation rejects tokens lacking a refresh token, failing loudly at setup
Health endpoint claimed a healthy token while every call failed	Health was computed from file age alone	Health now reflects the last real API outcome
Blank chart in the PWA	Canvas measured while hidden has clientWidth 0 → zero-width bitmap	Fallback width plus redraw on resize and visibility change

8.4 The token investigation — and a corrected diagnosis

Tokens were dying within hours instead of the documented seven days. The first two diagnoses were **wrong**, and the correction is the most instructive part of the project.

- **Hypothesis 1 (wrong):** a stale process presenting a superseded refresh token causes Schwab to revoke the family. Plausible, and it motivated real hardening — an in-lock staleness re-check, re-entrant locking, atomic writes — but it was not the cause.
- **Instrumentation instead of speculation.** Every token read and write was journaled with PID, command and a sha256 fingerprint of the refresh token.
- **What the journal proved:** the fingerprint was *identical on every line, including the WRITE* — **Schwab does not rotate the refresh token on refresh.** There was never a stale token to reuse, so the entire theory was impossible.
- **The real signature:** last successful refresh at 14:46, first failure 25.3 hours later, with **zero process events in between**. The token died while completely idle. No local race can explain that — it is server-side revocation, most likely a login started but never completed (approving the app re-authorizes it and can invalidate existing tokens even if the redirect URL is never pasted).
- **The tool's own verdict was also wrong** and had to be corrected: it flagged CONCURRENCY on two sequential reads, which cannot race when nothing rotates, and displayed journal times in ET against token timestamps in local time — making a re-login appear to precede the failures it fixed.

Lesson: when a system fails intermittently and two plausible theories have already been wrong, stop theorizing and instrument it. The diagnostic must also be validated — a confidently wrong diagnostic is worse than none.

9. Anticipated Questions

Prepared answers to the questions this project invites.

Q. Why compute gamma yourself instead of using the broker's greeks?

Three reasons. Auditability: I cannot verify a vendor's model, rate assumption, or dividend treatment, and each changes the number. Consistency: vendor greeks across strikes may be computed at different times or conventions. Control: recomputing lets me vary the rate, dividend yield, or exercise-style assumption and see the effect. Vendor IV is still an input, so this does not remove vendor dependence — it isolates it to one clearly labelled place.

Q. What is the biggest source of error?

The dealer sign convention. Assuming dealers are long calls and short puts is an inference about positioning, not measured data, and the flip's existence depends on it — flip the put sign and all gamma becomes positive, so the flip vanishes. That is why the tool reports a $\pm 50\%$ sensitivity band and prints LOW CONFIDENCE when the level moves more than 1% of spot. It can now also be tested empirically against CBOE open-close data, which converts the assumption into a measurement.

Q. Why run two profiles instead of one number?

Because they answer different questions and mixing them is a modeling error. The intraday profile covers 0DTE through the front week plus the next monthly OpEx, where essentially all real-time hedging pressure lives, and blends today's volume into stale open interest. The structural profile uses all expiries with open interest only and barely moves day to day — that stability is the point, since it locates multi-day walls. On a recent live run they disagreed on regime outright: intraday short gamma, structural long gamma. The tool flags divergence above 1% of spot, because it tells you the near-dated book is pulling the flip off the standing structure and it will snap back once that gamma expires.

Q. Why blend volume into open interest?

Open interest is published once daily, so on 0DTE it describes contracts that existed overnight, not what is being hedged now — it is stale by mid-morning and systematically understates the gamma driving the tape. I blend today's volume in at full weight for 0DTE, half through the front week, and not at all beyond. The honest caveat is that volume includes closing trades, so the blend is an upper bound on fresh positioning. I bias that way deliberately, because understating live 0DTE gamma is the larger error, and I print the uplift so the size of the correction is always visible.

Q. Why SPY instead of SPX?

Forced by the data source, and I verified it rather than assuming. Schwab returns zero open interest for cash-index options while SPY returns full OI on the identical request, so SPX GEX is not computable from this vendor at any price. SPY is the standard proxy; the tool cross-quotes levels to SPX using the live ratio and documents that it measures one underlying, not the full complex.

Q. How do you know your flip level is correct?

I test it against a closed form I derived by hand. For one call and one put with equal weight and matching vol and tenor, the common factors cancel at the zero, so the crossing must satisfy $S^* = \sqrt{K_c \cdot K_p} \cdot \exp(-cT)$. The grid search has to land on 97.5288 for my parameters. That validates the interpolation and crossing detection against mathematics rather than a recorded output.

Q. Why calendar time rather than trading time for ODTE?

Because I ingest vendor IV rather than deriving it. Gamma depends on sigma-squared times T, and the vendor annualizes IV on a calendar clock. Pairing that sigma with a 390×252 trading clock breaks the pairing and inflates ODTE gamma severalfold. Trading time is better only if you re-derive IV under the same clock. Given my inputs, calendar time is the internally consistent choice.

Q. Talk me through the hardest bug.

Tokens dying in hours instead of days. My first two diagnoses were wrong — I blamed a stale process presenting a superseded refresh token. So I stopped guessing and instrumented it: every token read and write journaled with PID and a hash of the refresh token. The journal showed the fingerprint was identical on every line including the write, which proved Schwab does not rotate refresh tokens at all — my theory was impossible. The real signature was a 25-hour gap between last success and first failure with zero process events in between: it died idle, which is server-side revocation. I also had to fix my own diagnostic, which was reporting a false positive and mixing timezones. The lesson is that a confidently wrong diagnostic is worse than none.

Q. How do you test code where the correct answer is not obvious?

I look for independently derivable truth: closed forms with parameters chosen so terms cancel, published textbook values like the Hull example, invariants that must hold such as the flip being independent of contract multiplier, and degenerate inputs with a known required output. For infrastructure I test observable behaviour — spawning a real subprocess to prove the file lock actually excludes another process, and asserting that raw secrets never appear in the log.

Q. What does this tool NOT tell you?

It does not capture dealer direction — that is assumed unless I supply open-close data. It does not model charm or vanna, which dominate into monthly OpEx roll-off. It does not measure the full index complex. And it is a regime classifier, not an entry trigger.

Q. What would you do with a better data budget?

First, CBOE open-close volume on an ongoing basis to ground the dealer sign empirically instead of assuming it — that attacks the single largest error source, and the ingestion path already exists. Second, real-time signed volume for intraday ODTE, which is strictly better than my volume-blend proxy. Third, a vendor with index and futures-options open interest so the full SPX complex could be aggregated on common notional.

Q. What would you do differently starting over?

Validate the data source against the actual requirement before building on it. I built for SPX and discovered only at live-testing time that the vendor has no index open interest, which invalidated the original target instrument — a ten-minute check of one field would have saved a migration. And I would instrument the token lifecycle from day one rather than after two wrong diagnoses.

10. Quick Reference

Key constants and defaults

Constant	Value	Meaning
DEFAULT_TICKER	SPY	Index options have no OI on this vendor
DEFAULT_MULTIPLIER	100	Standard equity/ETF and index multiplier

Constant	Value	Meaning
DEFAULT_RATE	0.0469	Risk-free rate; always printed, override with --rate
TICKER_DIV_YIELDS	SPY 0.012, QQQ 0.006	Auto-resolved dividend yields with printed provenance
DEFAULT_PRICE_RANGE	0.10	±10% flip-search window
DEFAULT_GRID_STEPS	1000	Flip-search grid resolution
FRONT_WEEK_DAYS	5	Intraday profile horizon (with next monthly OpEx)
VOL_WEIGHT_ODTE / FRONT	1.0 / 0.5	Volume blend weights by DTE tier
DAY_COUNT / EXPIRY_HOUR_ET	365 / 16:00 ET	Calendar ACT/365 to the cash close
T_FLOOR_SECONDS	300	Bounds the 1/sqrt(T) singularity at the close
MATERIAL_FLIP_MOVE	0.01	LOW CONFIDENCE threshold: 1% of spot
ITM_DEPTH_FILTER / SPREAD	0.05 / 0.25	Deep-ITM depth and stale-quote spread limits
FETCH_DEADLINE_S	90	Hard wall-clock cap on a chain fetch

Operating commands

```
# shell helpers (~/.zshrc) work from any directory
gex --ticker QQQ                # 0DTE + all expiries
gex --ticker QQQ --profiles      # intraday vs structural layers
gex --ticker QQQ --watch 60     # fast-cadence intraday recompute
gex --demo                      # offline, no credentials

gexserver                      # API + phone app on :8787
gexauth --manual               # weekly Schwab re-login
gexoi --ticker SPY             # d0I history from the chain archive
gexoi --ticker SPY --open-close f.csv # empirical dealer sign
gexps                          # what is holding the token?
gexaudit                       # why did the token die?

python3 -m pytest test_gex.py   # 53 tests, no network
```

Glossary

Term	Definition
Gamma (Γ)	Rate of change of delta with respect to spot; second derivative of option value
Dealer gamma exposure	Aggregate dollar gamma held by market makers; sets the direction of forced hedging
Gamma flip / zero gamma	Spot level where aggregate dealer gamma changes sign
Long gamma regime	Dealers sell rallies, buy dips — volatility-damping, mean-reverting

Term	Definition
Short gamma regime	Dealers buy rallies, sell dips — volatility-amplifying, trend-prone
Call wall / put wall	Strikes of max call-side / min put-side gamma; pin-resistance and support-then-accelerant
0DTE	Options expiring the same session; very large gamma that disappears at 16:00 ET
OpEx	Monthly options expiration (third Friday), holding the bulk of index open interest
Sticky strike	Assumption that each strike's implied vol stays fixed as spot moves
Charm / vanna	Second-order greeks: delta decay over time, and delta sensitivity to volatility
Opening ratio	dOI / volume — how much of a day's flow opened rather than churned